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RESEARCH-ARTICLE

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FRANCISCO RANGEL, Polytechnic University of Valencia, Valencia, Valencia, Spain

PAOLO ROSSO, Polytechnic University of Valencia, Valencia, Valencia, Spain

ANIS CHARFI, Carnegie Mellon University in Qatar, Doha, Qatar

WAJDI ZAGHOUANI, Hamad Bin Khalifa University, Doha, Qatar

BILAL GHANEM, Polytechnic University of Valencia, Valencia, Valencia, Spain

JAVIER SÁNCHEZ-JUNQUERA, Polytechnic University of Valencia, Valencia, Valencia, Spain

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Francisco Rangel
PRHLT Research Center,
Universitat Politècnica de València
Spain
fraranpa@prhlt.upv.es

Paolo Rosso
PRHLT Research Center,
Universitat Politècnica de València
Spain
prosso@dsic.upv.es

Anis Charfi
Carnegie Mellon University
Qatar
acharfi@andrew.cmu.edu

Wajdi Zaghouani
Hamad Bin Khalifa University
Qatar
wzaghouani@hbku.edu.qa

Bilal Ghanem
PRHLT Research Center,
Universitat Politècnica de València
Spain
bigha@doctor.upv.es

Javier Sánchez-Junquera
PRHLT Research Center,
Universitat Politècnica de València
Spain
juasanj3@doctor.upv.es

ABSTRACT

This paper summarises the Author Profiling and Deception Detection in Arabic (APDA) shared task at PAN@FIRE 2019. Two have been the main aims of this year's task: i) to profile the age, gender and native language of a Twitter user; ii) to determine whether an Arabic text is deceptive or not in two different genres: Twitter and news headlines. For this purpose we have created three corpora in Arabic. Altogether, the approaches of 13 participants are evaluated.

CCS CONCEPTS

• **Computing methodologies → Artificial intelligence**; **Natural language processing**.

KEYWORDS

author profiling, deception detection, Arabic, Twitter, FIRE

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1 INTRODUCTION

This year at FIRE[1] we have organised the Author Profiling and Deception Detection in Arabic (APDA)[2] shared task. In this paper, we describe the resources that we have created and made available to the research community[3], illustrate the obtained results and highlight the main achievements. The Author Profiling and Deception Detection in Arabic consists of two tasks. In the next section we will describe each of them.

1.1 Task 1. Author Profiling in Arabic Tweets

Author profiling distinguishes between classes of authors studying how language is shared by people. This helps in identifying profiling aspects such as age, gender, and language variety, among others. The focus of this task is to identify the age, gender, and language variety of Arabic Twitter users.

1.2 Task 2. Deception Detection in Arabic Texts

We can consider that a message is deceptive when it is intentionally written trying to sound authentic. The focus of the task is on deception detection in Arabic on two different genres: Twitter and news headlines.

2 CORPORA

We have created the following corpora: the Arap-Tweet corpus for author profiling, and the Qatar Twitter and Qatar News corpora for deception detection. We briefly describe them below.

2.1 Arap-Tweet

This corpus was developed at the Carnegie Mellon University Qatar (CMUQ) [7] with the aim at providing with a fine-grained annotated corpus in Arabic. It contains 15 dialectical varieties corresponding to 22 countries of the Arab League. For each variety, a total of 198 authors (150 for training, 48 for test) were annotated with age and gender, maintaining balance for both variables. The following groups were considered for the age annotation: Under 25, Between 25 and 34, and Above 35. For each author, more than 2,000 tweets were retrieved from her/his timeline. The included varieties are: Algeria, Egypt, Iraq, Kuwait, Lebabin Syria, Libya, Morocco, Oman, Palestine Jordan, Qatar, Saudi Arabia, Sudan, Tunisia, United Arab Emirates and Yemen. More information about this corpus is available in [8].

[1] http://fire.irsi.res.in/fire/2019
[2] https://www.autoritas.net/APDA/
[3] Following a methodology that accomplishes with the EU General Data Protection Regulation [4].

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2.2 The Qatar Twitter corpus

In the context of the ARAP project, we created the Qatar Twitter corpus by retrieving during 2017 and annotating[4] tweets referring to the Qatar Blockade and the Qatar World Cup. Statistics about this corpus are shown in Table 1. The number of tweets for the blockade topic is completely balanced between credible and non-credible classes. For the World Cup topic the corpus is almost balanced, with a slightly smaller amount of credible tweets (48% / 52%).

2.3 The Qatar News corpus

We also created the Qatar News corpus by retrieving and annotating short contents such as headlines and/or excerpts from well-known Arabic newsletters. Statistics on this second corpus can be seen in Table 1. The number of documents is almost balanced, with a slightly smaller amount of credible news (47% / 53%).

Table 1: Distribution of credible and non-credible tweets per topic in the Qatar Twitter and Qatar News corpora.

Corpus	Topic	Credible	Non Credible	Total
Qatar Twitter	Blockade	115	115	230
	World Cup	262	281	543
	Total	377	396	773
Qatar News		889	999	1,888

2.4 Performance Measures

In this section we describe the performance measures used for evaluating the systems in the different tasks.

2.5 Author Profiling

Since the data is completely balanced, the performance is evaluated by accuracy, following what has been done in the author profiling tasks at PAN@CLEF[5]. For each subtask (age, gender, language variety), we calculate individual accuracies. Systems are ranked by the joint accuracy (when age, gender and language variety are properly identified together).

2.6 Deception Detection

As in this case the data is slightly imbalanced, we measure the performance with the macro-averaged F-measure.

3 PARTICIPANTS' APPROACHES AND RESULTS

The participants have used different features to address the task, mainly: *i)* *n*-grams; *ii)* stylistic features; and *iii)* embeddings. With respect to machine learning algorithms, the most used one was Support Vector Machines. Nevertheless, a couple of participants approached the author profiling task with deep learning techniques. In such cases, they used BERT and LSTM respectively. According to the results, traditional approaches obtained better performances than deep learning ones.

Table 2: Best result per task.

Author Profiling (Accuracy)		
Gender	0.8194	MagdalenaYVino
Age	0.6250	Yutong [6]
Variety	0.9778	DBMS-KU [1]
Joint	0.4556	DBMS-KU [1]
Deception Detection (macro F-measure)		
News	0.7542	Nayel [3]
Twitter	0.8541	KCE_Dalab [2]
Avg.	0.8003	Nayel [3]

In case of author profiling, the best overall result (45.56% joint accuracy) has been achieved by DBMS-KU [1] with combinations of word *n*-grams, character *n*-grams, and function words to train Support Vector Machines. The best result for gender identification (81.94%) has been obtained by MagdalenaYVino, with a combination of words and emoticons 2-grams and 3-grams. In case of age identification, the best result has been achieved by Yutong [6] (62.50%) with a Logistic Regression classifier trained with a combination of word unigrams with character 2 to 5-grams. Finally, in regards of language variety identification, the best result (97.78%) has been achieved also by DBMS-KU.

Regarding deception detection, the best overall result (0.8003 macro F-measure) has been achieved by Nayel [3] with *n*-grams weighted with TF/IDF and Support Vector Machines. The best result on the Qatar News corpus (0.7542 macro F-measure) has been also obtained by Nayel, while the best result on the Qatar Twitter corpus (0.8541 Macro F-measure) has been obtained by KCE_Dalab [2], who approached the task with a combination of word and character *n*-grams and FastText embeddings to train a Support Vector Machine.

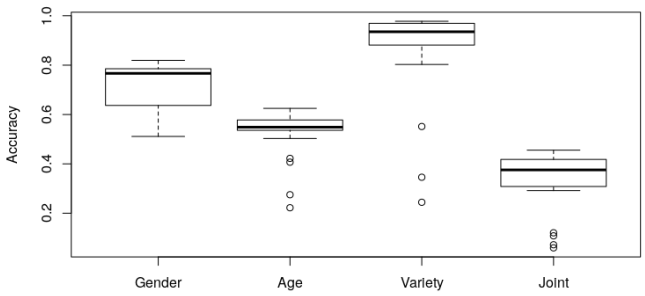

Figure 1: Distribution of results for the author profiling task.

Regarding author profiling, we can observe in Figure 1 that the highest results have been obtained in case of language variety identification, with most of the results very close to 1, although

[4]For both the Qatar Twitter and Qatar News corpora, the annotators were 20 students at the Hamad Bin Khalifa University, representing various Arab countries. The inter-annotator agreement was about 80%.
[5]https://pan.webis.de

with three outliers: two runs sent by Allaith (0.2444 and 0.3458), who did not send any description of their system, and the LSTM-based approach by Suman [5] (0.3458). We can also observe that the lowest sparsity occurs with age identification, where most of the systems obtained very similar results. In this case, there are also four outliers: the two systems of Suman (0.2222 and 0.2750) based on LSTM, and the two systems of Allaith (0.4069 and 0.4222). In case of gender identification, results are more sparse, but there are no outliers.

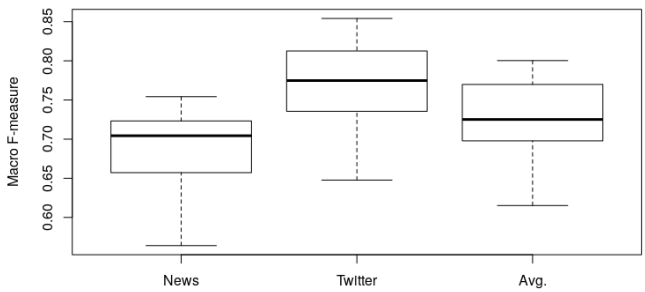

Figure 2: Distribution of results for the deception detection task.

In case of deception detection, we can observe in Figure 2 that the highest results have been obtained on the Twitter corpus, with similar sparsity on both genres. Perhaps, it should be highlighted that the distribution of results on the News corpus is more skewed to the right (the median is higher than the mean) and most systems close to the best performing ones.

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